

A positive-density family of dihedral fields with Bloch lattice index five

Abstract

We construct an infinite family of non-real dihedral extensions of degree ten over $\mathbb{Q}$, with fixed quadratic subfield $\mathbb{Q}(i)$, for which the Bloch lattice index Q_2 is five. A fixed Chebotarev set of rational primes has density $1/50$. For each parameter prime in this set, at least two of the four mixed cyclic quintic ray class extensions of conductor (25ℓ) have the asserted index. Their five-primary tame kernels are $(\mathbb{Z}/5)^2$. The proof combines a two-dimensional cup-product pencil, the first generalized Bockstein map, integral co-descent, and a motivic genus formula. It does not assume a Lichtenbaum special value formula or determine a Bloch lattice by numerical regulators.

1 The index and the statement

For a number field J , let $\mathcal{B}(J)$ denote the free quotient of its Bloch group, equivalently the free quotient of $K_3^{\text{ind}}(J)$. These identifications are natural in field embeddings; the regulator identifies the free quotient with its Bloch lattice. Let $M/\mathbb{Q}$ be a non-real D_{10} -extension, let k be its quadratic subfield, and let F_1, F_2 be two different quintic reflection subfields. The index considered by Browkin–Gangl and Guo–Qin is

$$Q_2(M) = [\mathcal{B}(M) : \mathcal{B}(k) + \mathcal{B}(F_1) + \mathcal{B}(F_2)],$$

where subgroups are understood via their natural images. Guo–Qin [2, Theorem 4.3] prove $Q_2(M) \mid 5$.

Put $k = \mathbb{Q}(i)$, $E = k(\mu_5) = \mathbb{Q}(\zeta_{20})$, and let S_0 consist of the primes above five and the infinite places. All sets of primes are pulled back to every extension in use. Write $K_2(O_J)(5)$ for the five-primary subgroup. The base vanishing needed below has a short verification.

Lemma 1. $K_2(\mathbb{Z}[i])(5) = 0$, and $H^2(G_{k,S_0}, \mathbb{F}_5(2)) = 0$.

Proof. The field $E = \mathbb{Q}(\zeta_{20})$ has class number one; the accompanying fixed-field certificate verifies this fact independently. Kummer theory gives $H^2(O_E[1/5], \mu_5) = \text{Br}(O_E[1/5])[5]$. The Brauer invariant sequence identifies this group with the kernel of $\mathbb{F}_5^2 \xrightarrow{+} \mathbb{F}_5$, one coordinate for each five-prime. Both primes are fixed by $\Delta = \text{Gal}(E/k)$, and local invariants show that Δ acts trivially on this kernel. Twisting by $\mathbb{F}_5(1)$ therefore gives a nontrivial cyclotomic character on $H^2(G_{E,S_0}, \mathbb{F}_5(2))$. Its Δ -invariants vanish. Since $|\Delta| = 4$, Hochschild–Serre identifies those invariants with $H^2(G_{k,S_0}, \mathbb{F}_5(2))$. The integral weight-two comparison with the tame kernel, reduction modulo five, and finiteness then give the asserted five-primary vanishing. $\square$

Theorem 2. *There is a set $\mathcal{P}$ of rational primes of natural and Dirichlet density $1/50$, all congruent to one modulo twenty, with the following property. For each $\ell \in \mathcal{P}$, at least two of the four anti-conjugate mixed cyclic quintic ray class extensions M/k of conductor (25ℓ) satisfy*

$$\text{Gal}(M/\mathbb{Q}) \simeq D_{10}, \quad Q_2(M) = 5, \quad K_2(O_M)(5) \simeq (\mathbb{Z}/5)^2.$$

Their discriminants are $\text{disc}(M) = -2^{10}5^{16}\ell^8$. If $a = v_5(\ell - 1)$ and $S = S_0 \cup \{v : v \mid \ell\}$, then

$$H^2(G_{M,S}, \mathbb{Z}_5(2)) \simeq (\mathbb{Z}/5^{a+1})^2.$$

In particular, there are infinitely many pairwise non-isomorphic such fields with the same quadratic subfield and the same degree over $\mathbb{Q}$.

The theorem permits a choice among four actual ray class fields for each parameter. It does not assert that an arbitrary preselected ray has index five. The density is a density of parameter primes, not a density ordered by field discriminant.

2 The fixed extensions and their Chebotarev set

The five-primary character space of the ray class group of k modulo (25) has dimension two. Complex conjugation exchanges the two primes above five. Its negative eigenspace has dimension one. Fix a nonzero character η_0 in this eigenspace, with fixed field M_0 . Thus M_0/k is cyclic of degree five and conductor (25), and $M_0/\mathbb{Q}$ is dihedral. This uses $h(k) = 1$ and $|\mu(k)| = 4$.

Let $T = \mathbb{F}_5(2)$. The integral group $H^1(G_{k,S_0}, \mathbb{Z}_5(2))$ is free of rank one over $\mathbb{Z}_5$: its rank is the Borel rank $r_2(k) = 1$, and its torsion is $H^0(k, \mathbb{Q}_5/\mathbb{Z}_5(2)) = 0$, since $\sqrt{5} \notin k$. The vanishing of the integral H^2 and Bockstein give

$$H^1(G_{k,S_0}, \mathbb{Z}_5(2))/5 \simeq H^1(G_{k,S_0}, T).$$

Restriction to E is injective, because $[E : k] = 4$ is prime to five. The Kummer comparison $H^1(E, T) = H^1(E, \mu_5) \otimes \mathbb{F}_5(1)$ turns its image into a one-dimensional nonzero subgroup $D \subset E^\times/E^{\times 5}$. This is the weight-two Tate line of [4]. It is unramified outside five in the sense that $E_T = E(\sqrt[5]{D})/E$ is unramified outside five.

Let $\Delta = \text{Gal}(E/k) \simeq C_4$, and let ω be its cyclotomic character modulo five. The line D has character ω^{-1} . Consequently, the Kummer pairing with target μ_5 shows that Δ acts on $\text{Gal}(E_T/E)$ with character ω^2 . On $\text{Gal}(M_0E/E)$, the action is trivial, since M_0E/k is abelian. Thus the two cyclic quintic extensions of E are different and linearly disjoint. The Tate line is also stable under conjugation over $\mathbb{Q}$; both factors, and hence

$$L = E_TM_0,$$

are normal over $\mathbb{Q}$. We have $[L : \mathbb{Q}] = 8 \cdot 25 = 200$.

Let $\mathcal{P}$ consist of the unramified rational primes whose Frobenius in $L/\mathbb{Q}$ is trivial on E_T and nontrivial on M_0E/E . In $\text{Gal}(L/E) = C_5 \times C_5$, this condition selects exactly four elements. It is invariant under conjugation, because both factors are normal over $\mathbb{Q}$. Chebotarev gives density $4/200 = 1/50$. Such primes split completely in E , so $\ell \equiv 1 \pmod{20}$. For either prime of k above ℓ , the unramified character η_0 has nonzero Frobenius value, and the Tate Frobenius is zero. Different choices of a prime of E change Frobenius by a nonzero scalar conjugation; its vanishing and the dimension of its span are unchanged.

3 A cup-product pencil on the fixed base field

Fix $\ell \in \mathcal{P}$, and let v_1, v_2 be the primes of k over ℓ . The anti-conjugate five-order character space modulo (ℓ) is one-dimensional. Choose a nonzero character η_ℓ there. For $\lambda \in \mathbb{F}_5^\times$, put

$$\eta_\lambda = \eta_0 + \lambda\eta_\ell, \quad M_\lambda = \bar{k}^{\ker \eta_\lambda}.$$

The two characters have different ramification sets and are independent. These are four different cyclic quintic layers, each fully ramified at both five-primes and both ℓ -primes and with conductor (25ℓ) . Conjugation negates each character, so each field is dihedral over $\mathbb{Q}$. The conductor–discriminant formula gives

$$\text{disc}(M_\lambda) = (-4)^5 N_{k/\mathbb{Q}}((25\ell)^4) = -2^{10} 5^{16} \ell^8.$$

Set $\mathcal{G} = G_{k,S}$, with $S = S_0 \cup \{v_1, v_2\}$. The base group $H^2(G_{k,S_0}, T)$ is zero. Localization and purity give

$$H^1(\mathcal{G}, T) \longrightarrow \bigoplus_{j=1}^2 H^0(\kappa(v_j), \mathbb{F}_5(1)) \longrightarrow 0, \quad (1)$$

$$H^2(\mathcal{G}, T) \xrightarrow{\sim} \bigoplus_{j=1}^2 H^2(k_{v_j}, T) \simeq \mathbb{F}_5^2. \quad (2)$$

In (1), each summand is $\mathbb{F}_5$, since $\ell \equiv 1 \pmod{5}$, and the map records tame inertia residues. Surjectivity follows from the preceding zero $H^2(G_{k,S_0}, T)$. For (2), localization identifies the new finite-field $H^1(\kappa(v_j), \mathbb{F}_5(1))$ with the local H^2 ; the next global H^3 is zero. The five-adic local H^2 groups vanish because those completions do not contain μ_5 .

Choose $y_1, y_2 \in H^1(\mathcal{G}, T)$ mapping to the standard basis in (1). The character η_0 is unramified at these primes, with Frobenius values $u, -u$, where $u \neq 0$. Locally, cup product pairs an unramified $\mathbb{F}_5$ -character perfectly with the one-dimensional tame inertia residue in $H^1(T)$; the cup of two unramified classes is zero. Consequently $\eta_0 \cup -$, restricted to the span of y_1, y_2 , has an invertible diagonal matrix C_0 in (2).

Let C_ℓ be the matrix for $\eta_\ell \cup -$ on the same two vectors. For η_λ , the matrix is

$$C_\lambda = C_0 + \lambda C_\ell.$$

Its determinant is a polynomial of degree at most two over $\mathbb{F}_5$, with nonzero constant coefficient. It has at most two roots. Therefore at least two of the four nonzero λ satisfy

$$\eta_\lambda \cup - : H^1(\mathcal{G}, T) \twoheadrightarrow H^2(\mathcal{G}, T). \quad (3)$$

No assertion about the distribution of residue symbols of varying prime generators is required: C_ℓ may depend arbitrarily on ℓ .

4 From cup product to integral co-descent

Fix a line satisfying (3). Write $M = M_\lambda$, $\mathcal{N} = \ker \eta_\lambda = G_{M,S}$, and $G = \mathcal{G}/\mathcal{N} = \langle \rho \rangle = C_5$. Put $\Omega = \mathbb{F}_5[G]$, $I = (\rho - 1)$, and $A = H^2(\mathcal{N}, T)$. Theorem 2.2.4 of Lam–Liu–Sharifi–Wake–Wang [5], with $R = \mathbb{F}_5$, $n = 1$, and finite $H = G$, gives

$$IA/I^2A \simeq \text{coker}(H^1(\mathcal{G}, T) \xrightarrow{\eta_\lambda \cup -} H^2(\mathcal{G}, T)).$$

All required flatness conditions hold over $\mathbb{F}_5$. For finite H , its Iwasawa cohomology is ordinary subgroup cohomology [5, Remark 2.2.1]; the first Bockstein is the cup product [5, Proposition 2.3.3]. Equation (3) gives $IA = I^2A$. Since $I^5 = 0$, it gives $IA = 0$. The natural top-degree co-descent isomorphism $A_G \simeq H^2(\mathcal{G}, T)$ therefore proves $A \simeq \mathbb{F}_5^2$.

Put

$$B = H^2(G_{M,S}, \mathbb{Z}_5(2)), \quad W = K_2(O_M)(5), \quad a = v_5(\ell - 1), \quad R = (\mathbb{Z}/5^a)^2.$$

These integral cohomology groups are finite. Cohomological dimension at most two and the Mittag-Leffler property for finite coefficient groups give continuous $H^3(G_{M,S}, \mathbb{Z}_5(2)) = 0$. Thus Bockstein gives $B/5B \simeq A$, and $d(B) = 2$.

The integral localization sequence is

$$0 \longrightarrow W \longrightarrow B \xrightarrow{\lambda_{\text{loc}}} R \longrightarrow 0. \quad (4)$$

The base group $H^2(G_{k,S}, \mathbb{Z}_5(2))$ is also R , because its tame kernel has zero five-primary part. Integral co-descent [3, Theorem 1.5 and the following paragraph] identifies $\text{cor} : B_G \xrightarrow{\sim} R$. The localization diagram is natural: on residue multiplicative groups the corestriction is the residue-field norm, while restriction is the ramification-index power. Here $f = 1$ and $e = 5$, so, in matched residue coordinates,

$$q := \text{cor} = \lambda_{\text{loc}}, \quad (\rho - 1)B = W, \quad \text{res } q = N_G. \quad (5)$$

The first equality uses the residue norm, not multiplication by the ramification index.

Lemma 3. *With the preceding notation, $W = 5^a B$, $5^{a+1}B = 0$, $N_G = 5 \text{id}_B$, and G acts trivially on W .*

Proof. A surjection from a finite abelian five-group with two generators onto $(\mathbb{Z}/5^a)^2$ forces both cyclic factors to have exponent at least a . Hence $|B/5^a B| = |R|$. Since $5^a B \subset \ker q$, comparison of orders in (4) gives $W = 5^a B$. Let $t = \rho - 1$. Then $tB = 5^a B$ and $t^j B = 5^{aj} B$. The identity

$$0 = 5t + 10t^2 + 10t^3 + 5t^4 + t^5$$

implies $5^{a+1}B \subset 5^{a+2}B$, as every other term has image contained in the latter group when $a \geq 1$. Finiteness forces $5^{a+1}B = 0$. Thus $t^2 B = 0$; in $N_G = 5 + 10t + 10t^2 + 5t^3 + t^4$, all nonconstant terms vanish, including $10tB \subset 5^{a+1}B$. Finally $tW = 5^a tB = 5^{2a}B = 0$. $\square$

5 The genus formula and the actual Bloch lattice

The motivic tame-kernel genus formula [4, Proposition 4.11 and Corollary 4.12] gives

$$\frac{|W_G|}{|K_2(O_k)(5)|} = 5^{2-t_2}.$$

Here t_2 is the dimension of the span of the two tame Frobenius elements in the fixed extension E_T/E . It depends only on the weight-two Tate line of k and the two tame ramified primes. All four mixed lines have the same tame set and nonzero local characters; changing λ multiplies a tame character by a nonzero scalar and does not change its norm kernel. By construction of $\mathcal{P}$, $t_2 = 0$ for all four lines.

Lemma 3 makes $W_G = W$. The genus formula gives $|W| = 25$, and $5W = 0$. Both cyclic factors of B must consequently have exponent $a + 1$. The equality $\text{res } q = 5 \text{id}_B$, with q surjective, gives $\text{res}(R) = 5B$. These groups have equal order, so restriction is injective. For a finite cyclic action $|B^G| = |B_G| = |R|$, so restriction is also surjective onto B^G . In particular the integral restriction

$$f_2 : H^2(G_{k,S}, \mathbb{Z}_5(2)) \longrightarrow B^G \quad (6)$$

has zero cokernel.

We include the lattice consequence to make the link to Q_2 explicit. It uses no special-value conjecture.

Proposition 4. *For a non-real D_{10} -extension $M/\mathbb{Q}$, if (6) has zero cokernel for a set S covering its relative ramification and all five-primes, then $Q_2(M) = 5$.*

Proof. Let $\Lambda = K_3^{\text{ind}}(M) \otimes \mathbb{Z}_5$. It is a free lattice of rank five. The torsion formula for indecomposable K_3 and the Chern comparison [9, Theorem 4.5 and Corollary 4.6] show that five-torsion would require $\mathbb{Q}(\sqrt{5}) \subset M$. But the unique quadratic subfield of a non-real dihedral field is imaginary. Thus there is no five-torsion. Levine's integral Galois descent [9, Theorem 4.13] identifies the actual subfield images with Λ^{C_5} and the two reflection invariant lattices. Its finite-quotient order condition is prime to the field characteristic, which is zero; it does not exclude degree five. The natural Suslin exact sequence identifies the free K_3 quotient with the free Bloch quotient, compatibly with these images.

Weight-two integral H^1 does not increase on enlarging S , because the possible new finite-field $H^0(\kappa(v), \mathbb{Z}_5(1))$ is zero. The comparison and [3, Theorem 1.5] give

$$\text{coker } f_2 = H^2(C_5, \Lambda) = \Lambda^{C_5}/N_G \Lambda = 0.$$

Write $\Lambda^{C_5} = \mathbb{Z}_5 y$, a saturated rank-one sublattice. Its quotient is a rank-one free module over $\mathbb{Z}_5[\zeta_5]$. Lift its generator to x , and write $N_G x = by$. The zero norm quotient forces b to be a unit; thus $\Lambda \simeq \mathbb{Z}_5[C_5]x$.

Let c be a reflection. On the invariant line it acts as -1 , since that line descends to the imaginary quadratic field, whose K_3 free quotient has no rational invariant part. In the regular group ring $\mathcal{R} = \mathbb{Z}_5[C_5]$, write $cx = wx$, so $w\bar{w} = 1$ and $\text{aug}(w) = -1$, with $\bar{\rho} = \rho^{-1}$. The unit $u = (1 - w)/2$ satisfies $w\bar{u} = -u$. Replacing x by ux normalizes the action to $c(rx) = -\bar{r}x$.

For a basis $e_j = \rho^j x$, a reflection $\rho^b c$ sends e_j to $-e_{b-j}$. Its invariant lattice is generated by the differences $e_j - e_{b-j}$. Two different reflections give a connected graph on the five indices: their composition is a nonzero translation. Their differences therefore generate the augmentation ideal $I_{\mathcal{R}}$. Adding $\Lambda^{C_5} = \mathbb{Z}_5 N_G x$, the quotient is $\mathcal{R}/(I_{\mathcal{R}} + \mathbb{Z}_5 N_G) \simeq \mathbb{Z}/5$. This computes the five-adic index of the actual subfield Bloch lattices. Guo-Qin's bound $Q_2 \mid 5$ makes it the full index. $\square$

Apply Proposition 4 to the good lines to finish Theorem 2. Varying ℓ changes the discriminant, so the result is an infinite family rather than a sequence of repetitions.

6 Fixed data and reproducibility

The theorem uses a fixed finite extension to define its prime set. It also has a small-field exact description. In $E = \mathbb{Q}(y)$, $y = \zeta_{20}$, one may take the Tate-line generator

$$g_2 = -5y^7 + 7y^6 - 3y^5 + y^4 - 5y^3 + 6y^2 - 3.$$

The supplied PARI/GP 2.15.5 certificate proves $h(E) = 1$, certifies the fundamental unit group, and records g_2 's coordinates $(1, 3, 1, 0)$. Thus it is not a fifth power. If $\sigma_{13}(y) = y^{13}$ and $c(y) = y^{-1}$, the unit coordinates of $\sigma_{13}(g_2)/g_2^2$ and $c(g_2)/g_2$ are respectively $(0, -5, -5, 15)$ and $(0, 0, 0, 10)$, where the final coordinate refers to the order-twenty roots of unity. Both ratios are fifth powers. This gives the required Δ -character ω^{-1} . Since it is an S_0 Kummer class and the base cohomology is one-dimensional, it generates the canonical line D .

The fixed pure-wild field M_0 is the splitting field of

$$f_0(X) = X^5 + 10X^3 - 10X^2 - 15X - 18.$$

A separate ray-class certificate over $\mathbb{Q}(i)$ proves its group, conductor, and discriminant. For $\ell \equiv 1 \pmod{20}$, membership in $\mathcal{P}$ can be checked by

$$g_2^{(\ell-1)/5} \equiv 1 \pmod{Q} \quad (Q \mid \ell \text{ in } E), \quad f_0 \bmod \ell \text{ irreducible.}$$

The second test detects the nontrivial cyclic Frobenius, since ℓ splits in the quadratic subfield. The exact certificate checks all eight residue symbols in E . It finds that 701, 941, 1321, 1381, 1801, 2081 meet both conditions, whereas 41 fails the Tate condition. These calculations check fixed data and examples; the infinite-family proof uses Chebotarev and the cup-product argument, not a finite search.

7 Relation to earlier work and scope

Browkin–Gangl [1, Section 12.4] already gave numerical evidence for $Q_2 = 5$, including a dihedral Hilbert-class-field example with quadratic subfield $\mathbb{Q}(\sqrt{-47})$. Their numerical lattice-generation qualifications must be kept distinct from an integral proof. Guo–Qin [2] established the dihedral divisibility theorem, which we use, and an infinite family with $Q_2 = 1$ and quadratic subfield $\mathbb{Q}(\sqrt{-3})$. General higher K -group index formulae and their cohomological interfaces already occur in Caputo [6, 8] and Bartel–de Smit [7]. The general integral co-descent and genus statements used here belong to Assim–Movahhedi [3, 4]; the generalized Bockstein theorem belongs to Lam–Liu–Sharifi–Wake–Wang [5].

The arithmetic construction here keeps the quadratic subfield fixed and selects among four mixed rays with a uniform two-root bound. It is not a new general Bockstein theory or a new polynomial root-counting lemma. Nor does it prove that every ray chosen by a separate Kummer algorithm is good. No claim of a formally verified proof, a first numerical example, or a named open problem’s historical attribution is made.

Computational and drafting assistance. Language-model assistance was used in developing the draft and accompanying exact-arithmetic scripts. The mathematical argument is supplied for independent human scrutiny. The artifacts contain exact checks, not Lean formalization or a certificate of novelty.

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